colloidal particles are separated on the filter and the impurities are removed as washings. This process is quickened by application of pressure. The separation of sol particles from electrolyte by filteration through an ultrafilter is called ultrafiltration. Collodion is 4% solution of nitrocellulose in a mixture of alcohol and water.

10.5.4 Properties of Colloids

1) Colour:

The colour of a sol is not always the same as the colour of the substance in the bulk. For example bluish tinge is given by diluted milk in reflected light and reddish tinge in transmitted light.

Colour of the sol, generally depends on the following factors.

(i) Method of preparation
(ii) Wavelength of source of light.
(iii) Size and shape of colloidal particle
(iv) whether the observer views the reflected light or transmitted light.

2) Size:

The size of colloidal particles ranges from 1nm (10^{-9}m) to 1000 nm (10^{-6}m) diameter.

3) Colloidal solutions are heterogeneous in nature having two distinct phases.

Though experiments like dialysis, ultrafiltration and ultracentrifuging clearly show the heterogeneous nature in the recent times colloidal solution are considered as border line cases.

4) Filtrability:

As the size of pores in ordinary filter paper are large the colloidal particles easily pass through the ordinary filter papers.

5) Non-Setting nature

Colloidal solutions are quite stable i.e. they are not affected by gravity.

6) Concentration and density

When the colloidal solution is dilute, it is stable. When the volume of medium is decreased coagulation occurs. Generally, density of sol decreases with decrease in the concentration.

7) Diffusability

Unlike true solution, colloids diffuse less readily through membranes.

8) Colligative properties

The colloidal solutions show colligative properties i.e. elevation of boiling point, depression in freezing point and osmotic pressure. Measurements of osmotic pressure is used to find molecular weight of colloidal particle.

9) Shape of colloidal particles

It is very interesting to know the various shapes of colloidal particles. Here are some examples

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